

1.2 From a To g And b-yond

The most common wireless technology is called Wi-Fi—short for Wireless Fidelity. This is actually a combination of different protocols that use the IEEE 802.11 specification standard. So, what is the IEEE 802.11 standard? In order to simplify things, detailed definitions are listed below.

IEEE

The Institute of Electrical and Electronics Engineers (IEEE) was formed on January 1, 1963, by way of a merger between the American Institute of Electrical Engineers (AIEE) and the Institute of Radio Engineers (IRE), both leaders in their respective fields.

The IEEE publishes 30 per cent of the world's literature relating to computer science, electrical and electronics engineering. The group has developed almost a thousand industry standards. Visit www.ieee.org for more information. For this book, we are only interested in a particular group of the IEEE.

802

IEEE 802 is a collection of networking standards that were designed or researched to run LANs and Metropolitan Area Networks (MANs). They are maintained by a specific committee called the LAN/MAN Standards Committee (LMSC). This committee oversees the functioning of smaller research groups. It is one of those groups—Group 11 to be precise—that we are getting to.

Group 11 (802.11)

A working group of the LMSC researched and developed the IEEE 802.11 WLAN standard, which is popularly called Wi-Fi today. This family of wireless standards currently includes four major wireless modulation techniques that all run on the same protocol, all developed by the IEEE 802.11 group. The standards are 802.11a, 802.11b, 802.11g, and 802.11n. The very first wireless standard was released in 1997, and was simply termed 802.11. It offered theoretical speeds of 1 Mbps and 2 Mbps, and operated at the 2.4 GHz